

Setting a knowledge boundary across teams: knowledge protection regulation for inter-team coordination and team performance

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Abstract

Purpose – As teams are built around specialized and different knowledge, they need to regulate their knowledge boundaries to exchange their specialized knowledge with other teams and to protect the value of such specialized knowledge. However, prior studies focus primarily on boundary spanning and imply that boundaries are obstacles to sharing knowledge. To fill this research gap, this study aims to indicate the importance of knowledge protection regulation, an activity that sets an adequate boundary for protecting knowledge, and investigate the factors that facilitate knowledge protection regulation and its consequences.

Design/methodology/approach – This study collected empirical data from 196 teams in seven organizations. Through a validation of the measurement model, data from 138 teams are used for further analysis. The hypotheses effects are assessed using a structural equation model.

Findings – The analysis results indicate that both task uncertainty and task interdependency enhance knowledge protection regulation in teams, and that information technology support moderates the relationship between task uncertainty and knowledge protection regulation. The results also indicate that knowledge protection regulation improves inter-team coordination and team performance.

Originality/value – This study focuses on knowledge protection regulation by adopting communication privacy management theory at the team level. The findings imply that boundary management is the process of communication and depends on the role the teams play in accomplishing their tasks. The findings also provide a new way to understand knowledge flow of the teams as well as the entire organization.

Keywords Knowledge boundary, Knowledge management across teams, Knowledge protection regulation

Paper type Research paper

Introduction

Companies have increasingly transformed themselves into team-based organizations to use the high information-processing capability resulting from the agility of small units (Galbraith and Lawler III, 1993; Sundstrom, 1999). As a subunit cooperating with other subunits within an organization, a team needs to develop boundaries that can resolve the uncertainty about its functions and expected outcomes and, simultaneously, allow outside resources and information to flow in (Hackman, 2002). In the absence of well-regulated boundaries, individuals' own rules rather than the shared rules of a team may be prioritized (Bechky, 2003), and individuals during cross-boundary interactions may struggle to find common ground in coordinating problems (Mors, 2010). In this sense, teams need to engage in boundary regulation that manages activities across boundaries (Faraj and Yan, 2009; Drach-Zahavy and Somech, 2010).

Received 17 April 2016
Revised 29 July 2016
Accepted 1 October 2016

Boundary regulation is a dynamic process as well as a set of rules and mechanisms for controlling the flow of resources (Altman, 1975; Petronio, 1991). Teams regulate their boundaries to perform the tasks that are directed outside the team to gain and protect resources for their own survival (Ancona and Caldwell, 1992; Yan and Louis, 1999). Among those resources, knowledge is particularly important (Kale *et al.*, 2002). Carlile (2002) suggests that knowledge should be localized, embedded and invested in different practices, and knowledge boundaries distinguish such specialized knowledge. Specialized knowledge here encompasses not only explicit knowledge, such as the codified results of different practices, but also tacit knowledge that cannot be separated from the people who perform those practices (Carlile, 2002). A knowledge boundary can be considered the border around the specialized knowledge domain of agents (Hawkins and Rezazade, 2012). Because the specialization of knowledge within boundaries has become the foundation on which an organization operates, setting knowledge boundaries is a critical challenge for coordinating knowledge across teams and maintaining it within teams (Carlile, 2002).

A team forms the smallest functional unit in which members work together on a permanent basis in an organization (Shin *et al.*, 2012). Most researchers have regarded a team as a work team comprising a recognizable group of members with stable, full-time and well-defined membership and who work on a unique task in a certain place for a certain period (Sundstrom *et al.*, 1990; Cohen and Bailey, 1997). Team members share and apply task-related knowledge to achieve tasks (Faraj and Sproull, 2000; Cannon-Bowers and Salas, 2001; Shin *et al.*, 2012). In this sense, this study focuses on the knowledge boundary corresponding to team boundaries (Argote *et al.*, 2003; Zellmer-Bruhn, 2003).

Because organizational knowledge resides in an organization's subunits under a team structure, the previous literature primarily investigates knowledge sharing across boundaries to enable knowledge to be used well beyond the team in which it is produced. Boundaries have been considered to hinder knowledge-sharing activities; thus, regulation should remove boundaries to enable knowledge to flow freely within an organization (Gladstein, 1984; Marrone, 2010; Hawkins and Rezazade, 2012).

Simply, opening knowledge boundaries is not always the best strategy (Eisenberg and Witten, 1987). Choi (2002) emphasizes a balance between external activities, such as managing interactions with other teams, and internal activities such as enforcing rules related to using internal resources. No *deus ex machina* exists for selecting between knowledge sharing and knowledge protection across boundaries (Carlile, 2004). Both opening a boundary (knowledge sharing) and closing a boundary (knowledge protection) improve a team's activities (Faraj and Yan, 2009). Teams need rules that govern the knowledge that stays within the boundary and the knowledge that enters and leaves it (Ratcheva, 2009). Applying the concept of emotion regulation in psychology (Gross, 1998), knowledge protection regulation is a set of rules for protecting knowledge boundaries and refers to the processes by which entities influence which, when and how they protect and disclose their knowledge. It helps to adequately manage a knowledge boundary from the perspective of closing a boundary (Manhart and Thalmann, 2015).

Prior studies on knowledge protection regulation are primarily conducted at the inter-organizational level. The study results suggest that organizations are faced with the challenge of balancing, "trying to learn" and "trying to protect" (Kale *et al.*, 2000), thereby making both knowledge sharing and knowledge protection critical (Trkman and Desouza, 2012). For example, unlimited openness of an organization's information to the outside endangers business plan implementation, particularly during a crisis (Eisenberg and Witten, 1987; Bok, 1989). Inappropriate knowledge sharing and the resulting knowledge loss in a competitive knowledge network can cause catastrophic results that erode core competencies in the long run (Hackney *et al.*, 2008; Gupta *et al.*, 2009).

The lack of research on knowledge protection regulation at the team level can be attributed to the fact that subunits in organizations share their goals and that their cooperative relationships are mandatory. Teams, however, cooperate and compete simultaneously with one another, which Tsai (2002) has called *coopetition*. In the same manner in which organizational boundaries distinguish organizations from their surroundings, internal boundaries can organize and coordinate various subunits within an organization (Rosenkopf and Nerkar, 2001). Knowledge boundaries stem from the coordination of problem-solving processes across subunits (Brown and Duguid, 2001). To resolve possible conflicts and to promote the interests of such subunits, each team not only needs knowledge provided by other teams but also must protect the knowledge that should be kept inside the boundary (van Knippenberg, 2003). Teams establish protective boundaries around themselves to concentrate on their tasks (Johnston and Linton, 2000).

The absence of a clear regulation for protective knowledge boundary causes confusion and conflict in identifying the ownership of knowledge and its exchange. Organizational policies do not entirely capture all the details of knowledge flow, particularly when teams need autonomy to better maneuver in a turbulent environment (Jones *et al.*, 2005). In addition to company policies, each team needs to manage its own processes and rules for adequate knowledge inflow and outflow.

To better understand knowledge protection regulation within teams, this study investigates the factors that affect this regulation and its association with team performance. Although originally developed for private information flow, the concept of an information boundary drawn from communication privacy management theory initiated by Petronio (1991, 2002) is useful to study the knowledge boundaries developed around teams. Communication privacy management theory suggests that boundary establishment is influenced by contextual and risk–benefit factors. In the team context, contextual and risk-benefit factors are related to task factors because teams are built on tasks (Stewart and Barrick, 2000). Information technology affects the manner in which information flows across teams (Alavi and Leidner, 2001; Bloodgood and Salisbury, 2001) and brings changes to the knowledge boundary and the nature of tasks (Howell and Cooke, 1989; Griffith *et al.*, 2003). The theory also alleviates tensions and clarifies communication to coordinate across teams. Therefore, on the basis of the theory, this study investigates how task-related context and risk–benefit affect the establishment of knowledge boundary regulation, how information technology (IT) support moderates this effect and how knowledge protection regulation can improve team performance.

Literature review and theoretical background

Knowledge protection regulation across teams

Team boundary regulation can be understood from team boundary work defined by Faraj and Yan (2009, p. 604) as “the activities that a team engages in to establish and maintain boundaries and manage interactions across those boundaries”. It does not only mean just the attainment of information and resources and the maintenance of relationships with the external environment, but also resource protection within a team (Ancona and Caldwell, 1988; Hansen, 1999; Reagans and Zuckerman, 2001). Because teams have various dependencies with their surrounding environments, their boundaries can be exposed to unwanted intrusion and uncertainties from various external interferences. The environment outside the boundaries is a source of disturbance (Yan and Louis, 1999; Thompson, 2011; Miller and Rice, 2013). Because external disturbances can harm teams’ internal operations, protecting teams from interference fosters the capability to manage their resources effectively (Faraj and Yan, 2009). The protection restricts the inflow of information and intrusions from external entities (Pawlowski and Robey, 2004), thereby preserving scarce internal resources and utilizing them for the team’s own purpose (Ancona and

Caldwell, 1988; Guinan *et al.*, 1998). Teams should be equipped with the appropriate processes to protect from external interferences (Scott, 1998).

Teams can attain organizational legitimacy for a certain degree of protection depending on circumstances and their capabilities (Ancona and Waller, 2007). Yan and Louis (1999) suggest that environmental forces, such as cross-functional team characteristics and information technology, affect teams' boundary activities. For example, Somech and Khalaili (2014) note that team heterogeneity negatively affects boundary-tightening activities for strengthened protection. Ancona and Caldwell (1988) show that software development teams' protection processes are regulated differently by an assessment of the legitimacy and expected cost of external requirements. When external requests require resources related to the core knowledge, knowledge protection is enforced and the requests can be denied (Mehta, 2006).

Some studies support the traditional view that a boundary is a hurdle for knowledge sharing and focus on the negative consequences of protection (Drach-Zahavy and Somech, 2010). This result needs to be further investigated because positive consequences of protection have been found. For example, Somech and Khalaili (2014) find that strengthened protection has positive effects on team innovation by encouraging team members to help one another and by providing constructive feedback. Protection can also contribute to maintaining a clear team identity by shielding team members from external threats or unwarranted influences (Ernst and Chrobot-Mason, 2011), and to maintaining psychological safety by establishing teams' reasons for being (Faraj and Yan, 2009).

The concept of boundary regulation: communication privacy management theory

Communication privacy management theory (Petronio, 1991, 2002) suggests the importance of a boundary, which is the border between the inside, where information is shared, and the outside, where information is not shared. A boundary's permeability ranges from thin to thick depending on the extent of the information that is allowed to pass through it. A thin boundary allows open access to inside information from the outside, whereas a thick boundary restricts information from the outside. Permeability is determined by the set of rules developed through five criteria: culture, gender, motivation, context and risk-benefit ratio.

Boundaries can become both personal and collective. When a boundary is collective, individuals inside it can share information but not with outside actors. A collective boundary can be established in two ways. First, it is built around a certain number of individuals regardless of any former shared membership. Second, a formal membership establishes boundaries. Individuals own the information, and the authorized possession of this information makes them members of the group. A group owns the information; thus, the affiliation to the group makes individuals legitimate members. In other words, although the individual is a formal member of the group and, thus, possesses certain information, from the moment that the individual loses affiliation to the group, he or she also loses ownership of that information.

When group members violate the rules of the group boundary, the boundary regulation fails and information unexpectedly leaks. However, the boundary regulation does not occur only through an intentional information breach but also through unclear rule establishment about the level of boundary permeability. Thus, developing clear rules is a precondition for information protection. A boundary regulation can alleviate the tension between protection and open availability of information, and communication becomes less confusing and unclear; thus, coordination can be enhanced.

By replacing the collective with team and private information with information that should be kept inside the boundary, communication privacy management theory provides a useful

lens for better understanding knowledge protection behavior across teams, particularly in the case of collective boundary.

Hypotheses and research model

Antecedents of knowledge protection regulation: task uncertainty and interdependency

Among the possible antecedents of knowledge protection regulation, we focus on: the boundary regulation factors drawn from communication privacy management theory and the factors that initially distinguish teams and become the basis of teams' operations.

First, by applying the theory to our study, for which the context is a team within an organization, we employ context and the risk–benefit ratio as the two most suitable antecedents of knowledge protection regulation. The effect of culture can be equivocal on all teams because it is shared by all of them. Gender and motivation are related to personal aspects; thus, the idiosyncratic characteristics of individuals rather than teams become salient. Context and the risk–benefit ratio are the best candidates for the antecedents of knowledge protection regulation because they are likely to affect the various teams differently but the same team equally.

Second, we focus on tasks as the determining factors that distinguish teams. Because teams are established on the basis of task allocation (Stewart and Barrick, 2000) and exist to achieve tasks, task-related factors are most important to a team's operation (Mathieu *et al.*, 2008). For example, prior studies suggest that different task characteristics influence the relationship among a team's inputs, processes and outcomes (Trist, 1981; McGrath, 1984; Goodman, 1986) and between team composition and team outcomes (LePine *et al.*, 1997; Neuman *et al.*, 1999), and a team's information-processing capability (Ancona and Caldwell, 1992).

Goodhue and Thompson (1995) submit two critical task characteristics, namely, task uncertainty and task interdependency. Task uncertainty determines the context in which teams operate with regard to unit structure and process (Drazin and Ven, 1985). When a task is interdependent with other teams, the risks and benefits are inherent in protecting or sharing knowledge (Lewicki *et al.*, 1998; Kankanhalli *et al.*, 2005). For example, interdependency makes it difficult to protect knowledge from other teams because doing so jeopardizes the reciprocal knowledge exchange relationship. Therefore, teams calculate the risks and benefits of regulating a protective knowledge boundary when task interdependency is high. This study submits task uncertainty as the task-related context factor and task interdependency as the task-related risk–benefit ratio factor.

The ambiguity from uncertain tasks hinders a shared understanding of task-related issues among team members (Galbraith, 1973; Van de Ven *et al.*, 1976). Because task uncertainty is “the difference between the information required to perform a task and the information already possessed” (Galbraith, 1973), individuals in a team seek information from outside regarding their tasks (Wood, 1986; Kramer, 1999) to narrow the knowledge gap (Souder and Moenaert, 1992). Accepting or disregarding received information should be firmly guided; otherwise, inappropriate information flow inadvertently increases uncertainty (Schulz, 2001). Because protected knowledge is related to core values, well-defined rules about knowledge protection can operate as basic guidance. The rules provide better comprehension about the possessed knowledge of a team (Ratcheva, 2009), protect the team against inappropriate inferences (Scott, 1998; Faraj and Yan, 2009) and minimize the likelihood of information misapplication (Lewicki *et al.*, 1998). In addition, they help confirm the information that can be shared (Majchrzak and Jarvenpaa, 2010), and the awareness of regulations can prevent inappropriate disclosure (D'Arcy *et al.*, 2009). For example, a software team performs a sentry process to tighten its boundaries to prevent undesired inputs from external entities, thereby avoiding their information-processing capacities from being overloaded (Jemison, 1984). This regulation also enables team members to work

with minimal external disturbance, save time and resources and better use their unique expertise and skills in innovative ways to achieve tasks (Yan and Louis, 1999). Hence, the following hypothesis is suggested:

H1a. Task uncertainty has a positive effect on knowledge protection regulation.

Task interdependency, which is defined as “the intensity and direction of a workflow relationship between two teams” (Gerwin and Moffat, 1997), becomes a constraint as team members rely on other teams’ knowledge to achieve their own tasks (Goodman, 1986). Task interdependency requires effective communication across teams to avoid mistakes, reworks and delays in allocating resources (Loch and Terwiesch, 1998; Kazanjian *et al.*, 2000). Therefore, teams under task interdependency make their boundaries thin to acquire and share knowledge (Joshi *et al.*, 2009; Marrone, 2010). Knowledge protection can give an impression to other teams that they are not willing to cooperate in sharing the necessary knowledge, thereby resulting in ineffective communication across teams (Mehta, 2006). Hence, we hypothesize that high task interdependency decreases the degree of knowledge protection regulation:

H1b. Task interdependency has a negative effect on knowledge protection regulation.

Moderating effect of IT support on task and knowledge protection regulation

IT support is “the degree of IT support for collaborative work, for communication, for searching and accessing, for simulation and prediction and for systematic storing” (Lee and Choi, 2003, p. 222). As teams’ knowledge work becomes more complicated, IT support becomes increasingly important in knowledge accumulation and utilization, and collaboration. Teams adopt numerous IT-based systems, such as e-mail, group support systems and electronic discussion forums, to search for task-related knowledge and to collaborate within and among teams (Orlikowski and Yates, 1994; Zigurs and Kozar, 1994; Alavi and Leidner, 2001). IT enables individuals to code and share best practices, such as tracking specific expertise, quick communication between them and the creation of networks (Alavi and Leidner, 2001; Bloodgood and Salisbury, 2001). Integrated IT systems can even replace existing interpersonal channels for knowledge exchange (Levina and Vaast, 2006).

IT has reduced teams’ communication costs (Schultze and Orlikowski, 2004) and helped them seek, interpret and use internal and external knowledge (Malone *et al.*, 1987; Schultze and Orlikowski, 2004). Well-developed IT can reconcile fragmented knowledge flows residing within a team (Gold *et al.*, 2001). Because knowledge protection regulation aims to establish an adequate level of knowledge flows across boundaries, IT support contributes to setting the right regulation level for boundary protection by helping to calculate the expected and required amount of knowledge flow. In this way, IT support affects the nature of tasks and their interrelationship (Howell and Cooke, 1989; Griffith *et al.*, 2003).

IT support facilitates teams’ efforts to standardize and integrate information flow (Morton and Hu, 2008), foster external knowledge exchange for additional perspective or clarification of perspectives on the task under non-routine task (Malhotra and Majchrzak, 2004) and identify and link related content in knowledge management activities (Zack, 1999). Thus, teams are more likely to recognize and find knowledge that is related to their core knowledge under well-supported IT. In addition, because IT enables the process to track the unwanted outflows of specific knowledge (Norman, 2001) and to control authentication (Manhart and Thalmann, 2015), the risk of knowledge leakage also decreases. IT support reduces information processing costs and increases information availability (Clemons *et al.*, 1993), decreases the tendency to regard external knowledge as the source of disturbances and alleviates the effort required for knowledge protection regulation. Therefore, the following hypothesis is suggested:

H2a. IT support weakens the positive relationship between task uncertainty and knowledge protection regulation.

IT support provides a set of arrangements that facilitate information exchange among teams (Argyres, 1999), and groupware improves the manner in which people collaborate (Olesen and Myers, 1999). IT support also provides easy access to information and visualization (Kautz and Thaysen, 2001), and its applications increase accuracy and expedite the flow of information (Gunasekaran *et al.*, 2004). Using visible information and connectedness with well-supported IT, the entities in a dyadic relationship can identify the knowledge of what they have and realize the benefit of information exchange. IT support can identify flow processes of knowledge through provided communication channels (Yeh *et al.*, 2006) and abolish barriers to communication among subunits in an organization (Lee and Choi, 2003). As teams under interdependent tasks may find greater benefits in loosening knowledge protection regulation using IT support, we suggest the following hypothesis:

H2b. IT support strengthens the negative relationship between task interdependency and knowledge protection regulation.

Consequences of knowledge protection regulation: coordination and performance

Because specialized knowledge is exchanged with other teams, an organization's subunits need to coordinate their use of knowledge (DeChurch and Mathieu, 2009). Knowledge protection regulation can help teams coordinate effectively by clarifying the knowledge that stays within a boundary. Such regulation prioritizes the objectives of knowledge management and specifies the role of the coordinated knowledge (Bloodgood and Salisbury, 2001). The regulation nurtures a safe and trusting environment by providing clear guidelines and enhancing coordination, such as joint problem solving and knowledge sharing (Faraj and Sproull, 2000; Faraj and Yan, 2009). With effective protection, knowledge sharing is more likely to occur within the right context (Randeree, 2006; Hurmelinna-Laukkanen, 2009). Building on prior work, we suggested the following hypothesis:

H3. Knowledge protection regulation has a positive effect on inter-team coordination.

Gold *et al.* (2001) submit that knowledge protection is an important element of the knowledge process capability that contributes to achieving organizational effectiveness. Well-established knowledge protection regulations are actually related to overall performance (Mills and Smith, 2011). Faraj and Yan (2009) show that avoiding information release – an activity they called boundary buffering – strengthens against external interference and creates a protected internal atmosphere for members, thus increasing team performance. Therefore, the following hypothesis is suggested:

H4. Knowledge protection regulation has a positive effect on team performance.

Prior work on the relationship between inter-team coordination and team performance is well established (Ruekert and Walker, 1987; Moenaert and Souder, 1990). Coordination ensures that appropriate resources are introduced at relevant times through communication and procedures (Holsapple *et al.*, 1996). Thus, coordination can enhance a team's performance (Ratcheva, 2009) and the quality of its outcomes (Hoegl *et al.*, 2004). We consider the overall established prior work and suggest the following hypothesis:

H5. Inter-team coordination has a positive effect on team performance.

Research method

Measurement

We adopted questionnaires from prior studies and modified their wording to reflect the team context of our study. Items for task uncertainty and task interdependency are from

Goodhue and Thompson (1995). The four IT support items are from Lee and Choi (2003). For knowledge protection regulation, we used four items applicable to the team context from 12 knowledge protection capability items in Gold *et al.* (2001). For inter-team coordination and team performance, we used the measure developed by Hoegl *et al.* (2004). All variables were measured using a Likert seven-point scale ranging from “strongly disagree” to “strongly agree”.

Data collection

We conducted a cross-sectional survey and collected data from full-time employees in seven Korean companies with assistance from their knowledge management teams. The industries in which these seven companies operate are: petrochemical, guarantee funds, system integration, construction, hospital, fine chemicals, and heavy industries. The petrochemical company manufactures petrochemical products through naphtha cracking and has approximately 900 employees. The company handling guarantee funds has 2,200 employees and is a quasi-government institute whose business includes credit guarantee, credit insurance and credit information management. The system integration company is an IT service company with approximately 7,000 employees. The construction company and hospital have approximately 400 and 1,000 employees, respectively. The fine chemical company, with 400 employees, manufactures fine chemicals and electronic materials. The heavy industry company, with more than 13,000 employees, engages in offshore plant building, shipbuilding and marine engineering.

Within these seven companies, the survey was distributed to 3,546 individuals working in 198 teams; 2,326 individuals from 196 teams responded to the questionnaires, for a response rate of 65.6 per cent. A team as the unit of analysis in this study is the smallest formal structure of each company and differs by well-defined tasks, such as inspection, technology planning, database management and customer support. The individual responses were aggregated into team-level variables with validity. The responses of individuals on teams with fewer than three responding members were eliminated to meet the minimum requirement for checking inter-rater reliability (James *et al.*, 1984) and the inter-class correlation coefficient (ICC) (Shrout and Fleiss, 1979). After removing these and incomplete responses, the responses of 1,542 individual team members on 138 teams were used for further analysis. Table I provides the demographic profile of the respondents.

Results

Measurement model

Before testing the measurement model, we checked the legitimacy of our sample for team-level analysis using inter-rater agreement (r_{wg}) (James *et al.*, 1984) and ICC (Shrout and Fleiss, 1979). The values exceeded 0.7 for both r_{wg} and ICC for all constructs, thereby satisfying the requirement for the aggregation of individual responses to team-level responses.

To test the measurement model, we first checked Cronbach's alpha for all constructs. As shown in Table II, Cronbach's alpha in all cases is greater than 0.9, indicating high reliability. We then performed confirmatory factor analysis with varimax rotation. The final items used for further analysis are presented in Appendix 1. As Table II indicates, all factor-loading scores are greater than 0.6 and all item loadings on their own constructs are higher than those of other constructs, suggesting that overall, convergent and discriminant validities of the measurement are met (Hair *et al.*, 2006).

Further, we examined the average variance extracted (AVE) and compared the square root of each construct's AVE to the correlation with other constructs. As displayed in Table III, AVE values are higher than 0.5 and the square root of the AVE value is greater than the

Table I Demographic profiles of respondents and teams

Characteristic	Frequency	(%)
<i>Individual (n = 1,542)</i>		
Gender		
Male	1,178	76.4
Female	364	23.6
Age		
20s	356	23.1
30s	563	36.5
40s	493	32.0
50s+	130	8.4
Tenure		
<2 years	548	35.5
2–5 years	342	22.2
5–10 years	314	20.4
>10 years	338	21.9
<i>Team (n = 138)</i>		
Team type		
Manufacturing	89	64.5
Non-manufacturing	49	35.5
Team size		
<10 members	46	33.3
10–20 members	41	29.7
20–30 members	14	10.2
30+ members	37	26.8
Number of teams in each company		
Petrochemical	11	8.0
Guarantee funds	5	3.6
System integration	10	7.2
Construction	31	22.5
Hospital	45	32.6
Fine chemical	15	10.9
Heavy industry	21	15.2

inter-construct correlations in all cases, indicating convergent and discriminant validity (Fornell and Larcker, 1981). Finally, we checked for the possible presence of common method bias by two methods. First, Harman's single-factor test (Podsakoff and Organ, 1986) indicated that the single factor explained only 22.10 per cent of the total variance; thus, no prevailing factor emerged from the test. Second, we decomposed the variance explained by the method factor and by the constructs, as suggested by Liang *et al.* (2007). The average substantively explained variance of the indicators is 0.928, whereas the average method-based variance is close to 0.021 (Appendix 2). Therefore, the results suggested that the presence of common method bias is not a serious problem for our study.

Structural model

To test the path analysis in our research model, the partial least square (PLS) method was used. The PLS method is a structure equation modeling method known to minimize prediction error using least square (Gefen and Straub, 2005) and an appropriate method for testing a small sample (Chin, 1998). As our sample size is relatively small ($n = 138$), PLS is a suitable method for our research model. The analysis results of our research model are depicted in Figure 1.

The hypothesized path from task uncertainty to knowledge protection regulation is significant, thus supporting *H1a*. However, task interdependency has a positive effect on knowledge protection regulation, the opposite of our expectation. Therefore, *H1b* is not supported. The moderating effect of IT support weakens the relationship between task uncertainty and knowledge protection regulation, thus supporting *H2a*. However, IT

Table II The results of the factor analysis ($n = 138$)

Item	Task uncertainty (TU)	Task interdependency (TI)	IT support (IT)	Knowledge protection regulation (KP)	Inter-team coordination (TC)	Team performance (TP)	Cronbach's alpha
TU1	<i>0.840</i>	0.288	0.055	0.143	0.053	0.197	0.939
TU2	<i>0.947</i>	0.112	0.069	0.081	0.048	0.044	
TU3	<i>0.922</i>	0.169	0.155	0.116	0.062	0.105	
TI1	0.280	<i>0.912</i>	0.138	0.172	−0.088	0.079	0.973
TI2	0.272	<i>0.916</i>	0.131	0.150	−0.096	0.074	
IT1	0.064	0.070	<i>0.842</i>	0.323	0.235	0.251	0.983
IT2	0.089	0.084	<i>0.823</i>	0.358	0.256	0.270	
IT3	0.136	0.150	<i>0.816</i>	0.335	0.218	0.284	
IT4	0.145	0.158	<i>0.824</i>	0.347	0.190	0.282	0.981
KP1	0.126	0.111	0.252	<i>0.875</i>	0.190	0.228	
KP2	0.101	0.112	0.335	<i>0.878</i>	0.170	0.194	
KP3	0.092	0.094	0.316	<i>0.873</i>	0.189	0.197	0.953
KP4	0.188	0.186	0.338	<i>0.800</i>	0.194	0.295	
TC1	0.138	0.071	0.419	0.306	<i>0.608</i>	0.452	
TC2	0.050	−0.024	0.102	0.172	<i>0.852</i>	0.269	0.972
TC3	0.012	−0.166	0.257	0.139	<i>0.813</i>	0.370	
TC4	0.026	−0.126	0.254	0.201	<i>0.781</i>	0.433	
TC5	0.171	0.039	0.436	0.304	<i>0.617</i>	0.476	0.972
TP1	0.185	−0.039	0.320	0.184	0.357	<i>0.757</i>	
TP2	0.119	0.080	0.226	0.250	0.280	<i>0.849</i>	
TP3	0.134	0.101	0.237	0.235	0.338	<i>0.826</i>	
TP4	0.061	0.053	0.137	0.138	0.240	<i>0.897</i>	
TP5	0.069	0.102	0.245	0.203	0.249	<i>0.855</i>	

Note: The numbers in italic are the item loadings on their own constructs

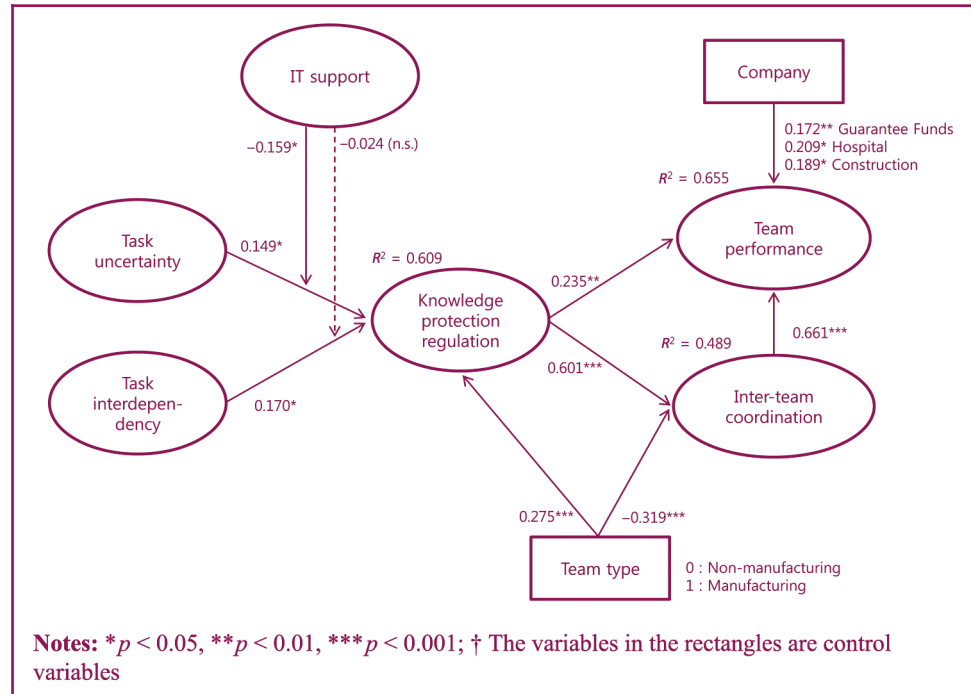
Table III Descriptive statistics and the correlation matrix for constructs

Construct	Mean	SD	TU	TI	IT	KP	TC	TP
Task uncertainty (TU)	5.011	0.772	0.943					
Task interdependency (TI)	5.158	0.946	0.497**	0.987				
IT support (IT)	4.594	0.852	0.310**	0.312**	0.976			
Knowledge protection regulation (KP)	4.607	0.961	0.334**	0.346**	0.718**	0.974		
Inter-team coordination (TC)	4.879	0.741	0.248**	0.040	0.678**	0.589**	0.920	
Team performance (TP)	5.201	0.759	0.312**	0.200**	0.624**	0.559**	0.778**	0.948

Notes: $n = 138$; The numbers in the diagonal are the square roots of the variance shared between the constructs and their measures. Off-diagonal elements are correlations among constructs; ** $p < 0.01$

support does not moderate the relationship between task interdependency and knowledge protection regulation, indicating that $H2b$ is not supported. As predicted, knowledge protection regulation affects inter-team coordination and team performance, thus supporting $H3$ and $H4$. Finally, inter-team coordination significantly affects team performance, which supports $H5$.

To control team and company types on knowledge protection regulation, inter-team coordination and team performance, we added team type, team size and company as control variables. The results showed that none of the team size effects were significant; the effect of team type on team performance was not significant and the effect of company was only significant on team performance. Regarding the company, the performance of the teams in four companies are indifferent but the performance of the teams in the guarantee fund company, the hospital and the construction company seem better than those in other companies. With regard to team type, the results show that manufacturing teams are more involved than non-manufacturing teams in knowledge protection regulation and less involved in inter-team coordination.

Figure 1 Research model results ($n = 138$)

Discussion

The results show that both task uncertainty and task interdependency enhance a team's knowledge protection regulation that leads to inter-team coordination and team performance. The test result of the path between task uncertainty and knowledge protection regulation is shown as expected; task uncertainty, which represents the context among the antecedents of the protective knowledge boundary suggested by communication privacy management theory, has a positive effect on knowledge protection regulation. In contrast, the test result of the relationship between task interdependency and knowledge protection regulation is contrary to our expectations; task interdependency has a positive rather than a negative effect on knowledge protection regulation. There are several possible explanations for this unexpected result. First, task interdependency, similar to task uncertainty, increases the need to clarify the knowledge that can be exchanged and the knowledge that should stay within the boundary. The clarification implies that teams assess the risks and benefits of a protective knowledge boundary and find that the benefits are greater than the risks. For example, the benefit of keeping knowledge from outside for competitiveness is greater than the risk of endangering reciprocal relationships with other teams. Second, psychologically safe teams tend to share more ideas (Edmondson, 1999). Working together under high task interdependency can diminish the psychologically safe environment for teams, which may thus feel a need to focus on protecting rather than sharing knowledge. The third possible explanation is that information useful to one party may turn out to be harmful to another party. Under the shared goal of benefitting each other, teams may become more cautious in releasing information and may build more protective boundaries (Lewicki *et al.*, 1998).

Our study result verifies that IT support weakens the link that task uncertainty leads to knowledge protection regulation. Because IT support fulfills the information needs of teams with respect to transparency and clarity through facilitating communication and searching relevant roles, external disturbances do not always have to be uncertain, and the tendency to protect knowledge can be alleviated. The insignificant moderating effect of IT support on

the relationship between task interdependency and knowledge protection regulation requires further interpretation. We anticipated that IT support would contribute to alleviating the tendency for knowledge protection because task interdependency is expected to decrease the need for knowledge protection, whereas IT support was expected to strengthen this likelihood by enabling better communication and collaboration through features of IT. However, the result indicates that a tendency exists for task interdependency to increase the need for knowledge protection, and IT support has no effect on this relationship. We speculate that the protective aspects of IT, such as security (Neville *et al.*, 2003), offset the effectiveness of knowledge sharing.

Our study results reveal that knowledge protection regulation affects inter-team coordination and both have positive impacts on team performance. A knowledge boundary actually nurtures inter-team coordination even though the concept of a boundary seems to hinder knowledge sharing, which is a prerequisite for inter-team coordination. Teams that harness their knowledge resources seem adept at coordinating collective knowledge because the inability to protect their own knowledge actually leads them to terminate the coordination relationship (Yang *et al.*, 2014). This positive effect is important because inter-team coordination also leads to increased team performance. Our result that knowledge protection regulation has a positive effect on team performance is in line with the results of prior studies at the organizational level (Gold *et al.*, 2001; Mills and Smith, 2011). This finding suggests that the clarification of important resources through a protective knowledge boundary is effective not only at the organizational level – at which related entities are interested in achieving their own goals – but also at the team level at which related entities are willing to be more cooperative.

The findings related to team size and team type – used as control variables – suggest additional insights. Manufacturing teams are focused on developing more protective knowledge boundaries. These teams are more likely to value protective knowledge boundaries because their boundaries are relatively clear as opposed to those of non-manufacturing teams with respect to the original ownership of resources.

Implications

Theoretical implications

We see three important implications for the theory. First, the most significant implication of our results is that knowledge protection regulation is important to harness team coordination and team performance. The literature on knowledge management research has mostly focused on a sharing perspective and regards a knowledge boundary as an obstacle to knowledge sharing (Hawkins and Rezazade, 2012). Using field data, this study provides empirical evidence that a knowledge boundary actually benefits not only teams but also the entire organization. Our findings contribute to the advancement of the knowledge management literature by providing a different way to understand knowledge flow.

Second, we adopt the concept of a boundary from communication privacy management theory and provide a firmer theoretical ground for examining boundary regulation and its antecedents and consequences. Our result hints that boundary management is a communication process, and both the border of the boundary and its thickness vary depending on communication needs. This perspective is particularly applicable to the exploration of knowledge management because task performance through knowledge flow is the result of constant communication. Therefore, our study provides a better understanding of how teams align themselves in an environment in which they are intertwined with other teams through knowledge exchange.

Finally, our findings reveal the circumstances in which teams regulate knowledge boundaries. Task characteristics represented as task uncertainty and task

interdependency and team characteristics, such as team type, affect knowledge protection regulation. Because these characteristics are the foundation of a team's operation, our study shows that knowledge protection regulation is closely related to the innate nature of a team. Furthermore, our study reveals the effect of IT support on the relationship between knowledge protection regulation and task characteristics and strengthens the view that IT plays a critical role in innovating the manner in which the work is performed.

Practical implications

The findings of this study remind managers of the importance of knowledge protection regulation. A company is a unit that works as a large organism; each subunit in it should cooperate with other subunits as if each is an organ working as part of the body in which it is contained. From this perspective, knowledge in an organization should flow similar to the flow of blood: it circulates throughout the organization. In contrast, our result demonstrates that keeping knowledge within a team can actually encourage inter-team coordination and unleash team performance. Teams are not only the dependent parts of an entire organization but also are established and independent entities that pursue their own effectiveness. To ensure such effectiveness of knowledge boundary, teams distinguish between resources to be shared and those that should stay. Managers should understand the roles and needs of teams, encourage them to clarify their own knowledge boundaries and assimilate them to benefit the entire organization. This may be done by different approaches, depending on the type of knowledge. In the case of explicit knowledge that can be exchanged through knowledge management systems, that approach should be to design well-articulated authorization options in a system that can be controlled and changed at a team's discretion. In the case of tacit knowledge, clear communication about the policies, rules and possible violations of boundary regulation with detailed examples should be established to guide the actions of individuals in a team.

As part of this assimilation, managers should also understand the nature of tasks that tend to establish more protective knowledge boundaries. Task uncertainty and task interdependency lead to more protective knowledge boundaries. If managers want organizational knowledge to flow freely across teams, they need to provide clear guidelines for performing tasks and cooperation. A possible solution is to identify the needs of knowledge protection regulation on the basis of their tasks and determine the extent of the IT support. If forcefully loosened, knowledge protection regulation only aggravates team performance, particularly because knowledge protection regulation is critical in nurturing inter-team coordination.

Lastly, an organization should understand that teams need to regulate knowledge boundaries. An organization that pursues a greater knowledge flow across teams should enable them to better process external knowledge with appropriate IT support. In contrast, an organization that aims to maximize team performance should encourage teams to develop clear knowledge boundaries on their own.

Limitations and future study

This study has several limitations. First, we collected data from seven companies in various industries and merged them to test the research model to increase generalizability; however, in this process, we may have compromised companies' idiosyncratic characteristics. Second, we selected task-related constructs to represent context and risk–benefit factors on the basis of communication privacy management theory. Constructs other than these could be selected to better represent the theory. Future research can use our findings as a starting point and extend the set of antecedents for knowledge boundary regulation. Third, because the primary objective is to find the antecedents and consequences of knowledge protection regulation, our study focuses on the thick knowledge boundary that makes it difficult to pass inside knowledge to the outside.

Because a team needs both a thick and a thin knowledge boundary to work as a subunit in an organization, future research can further pursue this issue by examining how teams build both boundaries. Such research will provide even richer insights when it is related to organizational culture and structure. For example, a stiffened organizational culture or centralized structure may lead to thick and inflexible boundaries. Therefore, additional research into the effects of the types of culture and the degree of centralization on boundary regulation efforts and forms will shed new light on how an organization evolves with certain boundary regulation processes.

Conclusion

This study adopted communication privacy management theory to explore the antecedents and consequences of knowledge protection regulation. The results of this analysis verify that task uncertainty and interdependency have positive effects on knowledge protection regulation and that IT support weakens the relationship between task uncertainty and protection regulation. The results also show that knowledge protection regulation has a positive effect on coordination and leads to higher team performance directly and indirectly via coordination. The findings demonstrate the benefits of regulating protective knowledge boundaries and suggest that knowledge boundary management should be not just a simple process of control but a communication process based on better understanding of team characteristics, such as its tasks and environment. The findings of this study extend our understanding of how knowledge boundaries function in an organization, showing that they not only hinder knowledge sharing but also enhance coordination and performance and therefore require regulating an optimal level of both knowledge protection and knowledge sharing.

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Appendix 1

Table A1 Measurement		
Construct	Measurement items	Source
Task uncertainty	Our team frequently deals with ill-defined business problems Our team frequently deals with ad-hoc, non-routine business problems Frequently, the business problems our team works on involve answering questions that previously were never asked in quite that form	Based on Goodhue and Thompson (1995)
Task interdependency	The business problems that our team deals with frequently involve more than one team The problems that our team deals with frequently involve more than one team	Based on Goodhue and Thompson (1995)
IT support	Our team is provided with IT support for collaborative works regardless of time and place Our team is provided with IT support for communication among team members Our team is provided with IT support for searching for and accessing necessary information Our team is provided with IT support for systematic storing	Based on Lee and Choi (2003)
Knowledge protection regulation	Our team has processes to protect knowledge from inappropriate use outside the team Our team has processes to protect knowledge from theft outside the team Our team has technology that restricts access to some sources of knowledge Our team clearly communicates the importance of protecting knowledge	Based on Gold <i>et al.</i> (2001)
Inter-team coordination	Connected processes and activities are well coordinated with other teams Duplicated and overlapping coordination activities are avoided We have no problems in coordinating with other teams Conflicts with other teams are settled quickly Discussions with other teams are conducted constructively	Based on Hoegl <i>et al.</i> (2004)
Team performance	So far, all team goals have been achieved The team's output so far is of high quality The team is satisfied with its performance to this point The project leadership can be fully satisfied with the task progress of our team	Based on Hoegl <i>et al.</i> (2004)

Appendix 2

Table AII Common method bias analysis

Construct	Indicator	Factor loading on construct (R1)	R1 ²	Factor loading on method (R2)	R2 ²
Task uncertainty	TU1	0.880***	0.774	0.083	0.007
	TU2	0.997***	0.994	−0.095*	0.009
	TU3	0.954***	0.910	0.013	0.000
Task interdependency	TI1	0.982***	0.964	0.013	0.000
	TI2	0.992***	0.984	−0.0136	0.000
IT support	IT1	1.014***	1.028	−0.061	0.004
	IT2	0.969***	0.939	0.014	0.000
	IT3	0.953***	0.908	0.028	0.001
	IT4	0.967***	0.935	0.018	0.000
Knowledge protection regulation	KP1	0.983***	0.966	−0.016	0.000
	KP2	1.056***	1.115	−0.086**	0.007
	KP3	1.035***	1.071	−0.073	0.005
Inter-team coordination	KP4	0.824***	0.679	0.172***	0.030
	TC1	0.671***	0.450	0.289***	0.084
	TC2	1.031***	1.063	−0.209*	0.044
	TC3	1.181***	1.395	−0.294***	0.086
	TC4	1.072***	1.149	−0.140**	0.020
	TC5	0.672***	0.452	0.318***	0.101
Team performance	TP1	0.884***	0.781	0.044	0.002
	TP2	0.913***	0.834	0.061*	0.004
	TP3	0.852***	0.726	0.129***	0.017
	TP4	1.143***	1.306	−0.233	0.054
	TP5	0.955***	0.912	−0.008	0.000
Average			0.928		0.021

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

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